

Aliah University
BCA 2nd Semester-Spring 2024
Python Programming [BCAUGSEC1201]
Full marks - 80 Time- 3 hrs.

Group A

1. Suppose a list is [4, 2, 2, 4, 0], Which of the following is the correct syntax for slicing operation?
a) `print(list1[2:])`
b) `print(list1[:2])`
c) `print(list1[:-2])`
☒ d) all of the mentioned
2. How can you concatenate two strings in Python?
a) Using the `concat()` method
b) Using the `join()` method
☒ c) Using the `+` operator
d) Any of the above
3. Output of the string = "Hello, World!" - `print(string[3:7])` is
a) "Hello"
b) "lo, "
☒ c) "lo, W"
d) "lo, World"
4. What is an object in OOP?
a) A variable
b) A block of code
☒ c) An instance of a class
d) A built-in function
5. What is the process of defining a new class based on an existing class called?
a) Polymorphism
☒ b) Inheritance
c) Abstraction
d) Encapsulation
6. What is the output of string = "Hello, World!" - `print(string.find("o"))`
☒ a) 4
b) 6
c) 7
d) -1
7. What is the output of string = "Hello, World!" - `print(len(string))`
a) 12
☒ b) 13
c) 11
d) 10
8. Replacing a method in the superclass with a new implementation in the subclass is known as
a) method overriding
b) method overloading
c) both
d) none
9. Which character is used in Python to make a single line comment?

- a) /
 - b) //
 - c) #
 - d) !
10. What will be the result of `my_list = [1, 2, 3, 4, 5] - my_list.remove(3)`
- a) [1, 2, 4, 5]
 - b) [1, 2, 3, 4]
 - c) [1, 2, 4]
 - d) [1, 2, 5]

Group B [4*5]

Answer any four

- ✓ 11. Explain with an example how functions can be created in Python, how parameters can be passed and returned, and how to use a function.
- ✓ 12. Write a program to find the factorial of a given number.
- 13. Write code snippet for creating a folder, storing a text file inside, write 3 lines of text for your home address, and then read the lines sequentially from the file and print it.
- ✓ 14. Write a program to print the first n numbers of the Fibonacci series.
- 15. Explain two different ways of creating a loop in Python.

Group C [5*10]

Answer any five

- ✓ 16. Create a Python class to implement stack operations. The class should contain two data variables: a list (data) and an integer (top), and two methods push() and pop(). [10]
- 17. Write a program to i) find the sum of all odd numbers till 100. ii) find the product of digits of a number. [5+5]
- 18. Write a program to i) to declare and initialize a numpy array with 1 to 10 in reverse order. ii) Store the last 4 elements in an another array. iii) Check if a given number is palindrome or not [2+2+6]
- ✓ 19. List different operators in Python in the order of their precedence. Differentiate between if-else and elif statement [5+5]
- ✓ 20. Write the difference between a list and an array with examples. Reverse a Python string using loop [5+5]
- 21. Write a program to input two matrix, multiply them and print the result. [10]
- ✓ 22. Illustrate the four prime features of Object Oriented Programming with suitable examples. [10]

Job is full